

NAUTRACE PROJECT HANDBOOK

Samundar Ka CID & Forensic Detective — Poore Project Ko Aasan Bhasha Me Samjho

Yeh official handbook hamari 6-member team ke liye banaya gaya hai taaki har member project ke working mechanics, real-life problem, physics equations aur jury defense playbook ko simple aur clear Hinglish me samajh sake.

Core Problem:	Nearest-Ship Fallacy (Oil drifts 30-50 km under wind & ocean currents before satellite observation)
Our Solution:	Lagrangian Backward Hindcasting (RK4) + Mahalanobis Origin Ellipses + Explainable Attribution
Legal Moat:	Explicit UNKNOWN_NON_AIS Hypothesis (IMO Resolution A.1106 compliant) + SHA-256 Chain of Custody
Live Interface:	Desktop Forensic Workspace with 4-Hour Timeline Scrubber & Convex AI Report Synthesis

1. Pehle Ek Real-Life Kahani Samjho (Samundar Ka Hit-and-Run)

Nadi Aur Pul Ki Kahani:

Maan lijiye ek behne wali tez nadi (river) hai. Ek aadmi pul (bridge) par khada hota hai aur nadi me ek tel ka dabba fenk kar wahan se bhaag jata hai.

3 ghante baad police wahan aati hai aur dekhti hai ki pul ke 10 kilometer aage nadi me tel beh raha hai. Aur theek us tel ke bagal me ek innocent naav wala machhli pakad raha hai.

Agar police us machhli pakadne wale ko pakad kar jail me daal de ki *'Tel ke paas to tum hi khade the, tumne hi fenka hai!'* — to yeh sarasar galat aur be-insaafi hogi! Kyunki tel to 3 ghante pehle 10 kilometer peeche pul se fenka gaya tha!

Yahi Sabse Badi Problem Hoti Hai Samundar Me:

Bade-bade cargo ships aur oil tankers apna engine ka ganda oily kachra (bilge waste) samundar me chupke se dump kar dete hain port cleaning ke laakhon रुपये bachane ke liye. Jab satellite space se photo leta hai, tab tak hawa aur leharon ke bahav se wo tel 30 se 50 kilometer aage beh chuka hota hai.

Nautrace Ka Asli Kaam: Nautrace samundar ka ek **Digital Forensic Detective (CID / CBI)** hai. Yeh satellite ki photo dekh kar time ko rewind (ulta) chalata hai, pata lagata hai ki tel kahan se behna shuru hua tha, aur us time par wahan se kaunsa jahaaz guzar raha tha.

2. Guide Page Ke 6 Tabs Ka Poora Khulasa

Tab 1: The Problem We Solve (Asli Dikkat Kya Hai?)

- **Baaki projects kya galti karte hain:** College aur hackathons me log simple AI bana dete hain jo photo me tel dekhta hai aur bagal wale jahaaz ko mujrim bol deta hai.
- **Asliyat:** Samundar me tel rukta nahi hai, beh jata hai. Pass wala jahaaz aksar innocent hota hai.
- **Nautrace kya alag karta hai:** Hum physics aur samundari leharon ke hisaab se tel ko ulta chala kar dekhte hain (Hindcasting) ki 2-4 ghante pehle wo tel kahan tha.

Tab 2: End-to-End Pipeline (8 Aasan Steps)

Step 1:	Sentinel-1 Satellite Photo: Radar camera jo badal aur raat ke andhere me bhi samundar ki photo leta hai.
Step 2:	Look-Alike Filter: Hawa band hone par shant paani kaala dikhta hai. System pehle hawa check karta hai.
Step 3:	Metocean Data: Europe ke ocean models se paani aur hawa ke bahav ki speed aur direction lena.
Step 4:	RK4 Hindcast (Rewind): 500 virtual tel ke kankaron ko computer me minute-by-minute ulta chalana.
Step 5:	Crime Zone Ghera: 50% core zone (Yellow) aur 90% outer search zone (Cyan) Mahalanobis ellipses banana.
Step 6:	AIS Trajectory Reconstruction: Samundar ke sabhi jahaazon ke GPS data ko clean karke match karna.
Step 7:	Scoring Engine: Maths calculate karta hai kis jahaaz ka track aur time match hua. Sab door ho to Unknown jeetta hai.
Step 8:	Forensic Report: Convex AI dwara SHA-256 digital signature aur proof ke saath court-ready legal dossier.

Tab 3: Mathematics & Physics (Bina Darr Ke Samjho)

- **Particle Motion (dx/dt):** Tel ki raftaar = (Samundar ka paani) + (Leharon ka dhakka) + (Hawa ka lagbhag 3% dhakka).
- **Mahalanobis Ellipse (Anda-aakar Ghera):** Kyunki hawa aur lehar ek specific disha me beh rahi hoti hain, isliye shaq ka daayra gol nahi, leharon ki disha me lamba banta hai.
- **Interactive Simulator:** 1.3 km paas = Score 0.79 (Top Suspect); 10 km door = Score 0.15 (Ruled Out) aur Unknown source jeet jata hai.

Tab 4: 7 Explainability Scores (CBI Ki 7 Kasam)

Metric Name	Asli Matlab	Tanker Score
1. Distance (Spatial)	Kya jahaaz us point ke kitna paas se guzra?	92/100 (1.3 km)
2. Time (Temporal)	Kya usi ghante me guzra jab tel dump hua tha?	86/100 (11 min)
3. Heading (Disha)	Kya jahaaz usi disha me ja raha tha jis line me tel baha?	73/100
4. Envelope Overlap	Kya jahaaz ne 50% ya 90% shaq wale ghere ko cross kiya?	90/100
5. GPS Continuity	Kya jahaaz ka GPS continuous tha ya beech me band hua?	98/100
6. Speed Anomaly	Kya achanak jahaaz slow hua tha? (Dump speed drop)	40/100
7. Rank Stability	Hawa-paani 15% hilane par bhi kya yahi jahaaz #1 rehta hai?	88/100

Tab 5: Real-World Issues & Hard Truths (Asliyat Me Kahan Faste Hain?)

- 1. **Hawa Kam Hona (< 3 m/s):** Hawa band hone par paani shant ho jata hai aur satellite dhokha kha sakti hai. Nautrace pehle weather data se hawa check karta hai.
- 2. **Tel Girne Ka Exact Time Kisi Ko Nahi Pata:** Satellite bas photo leti hai. Tel 2 ghante pehle gira ya 6 ghante pehle, koi nahi jaanta. Isliye hum ek single line nahi, poori time-window check karte hain.
- 3. **Chalu Captains (GPS Band Karna / Dark Vessels):** Tel dump karte waqt jahaaz GPS (AIS) 15 minute ke liye switch off kar dete hain. Nautrace be-kasoor par ilzaam lagane ke bajaye bolta hai: **'UNKNOWN / Non-AIS Source (Score 0.86)'** (IMO Resolution A.1106 compliant).

Tab 6: 5-Minute SIH Jury Playbook (Judges Ke Saamne Kya Bolna Hai?)

Minute 1: Problem Statement & Trap

"Respected Jury: Baaki log satellite photo me tel dekh kar uske bagal wale jahaaz ko mujrim bol dete hain. Par samundar me tel hawa aur leharon se ghanton me 40 km beh jata hai. Isliye pass wala jahaaz innocent hota hai. Humne ek real forensic pipeline banayi hai."

Minute 2: Physics Hindcasting Dikhao

"Screen par dekhiye: Humne 500 virtual particles ko RK4 physics aur Copernicus ocean currents ke zariye ulta rewind kiya hai, jisse 50% aur 90% shaq ka ghera nikalta hai."

Minute 3: Timeline Scrubber Chalao

"Jaise hi scrubber ko 02:50 UTC par le jaate hain, aap dekhenge ki Container Ship to abhi paas aaya hai, par Tanker 2 ghante pehle theek usi ghere se guzra tha jahan se tel nikla tha."

Minute 4: Unknown Hypothesis Ka Jadoo

"Humne IMO Resolution A.1106 follow kiya hai. Agar jahaaz ne GPS band kiya tha, to humara model kisi innocent par ilzaam nahi lagata, Unknown source ko rank #1 deta hai."

Minute 5: Court Dossier Export Karo

"Hum [Generate AI Brief] dabate hain. Convex AI court me pesh karne layak formal English report banata hai, jisme SHA-256 digital signatures lagte hain jo legally valid hain."

3. Quick Reference Cheat Sheet

Component	Bilkul Aasan Matlab
Sentinel-1 SAR	Aasman ka radar camera jo badal aur raat ke andhere me bhi tel pakadta hai.
CMEMS SMOC	Samundar ke paani ke bahaav aur leharon ka government ocean data.

RK4 Hindcast	Tel ke bahav ko rewind (ulta) karne wali physics calculation.
Mahalanobis Ellipse	Wo anda-aakar ghera jahan tel dump hone ka sabse zyada shaq hai.
AIS Data	Samundari jahaazon ki GPS location aur raftaar ki history.
Unknown Hypothesis	GPS band karke tel dump karne wale chalu jahaazon ka safety check.
Convex AI	Court me pesh karne layak formal English forensic report banane wala engine.